Adam Ormondroyd

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Education

University of Cambridge: Natural Sciences M.Sci. (Physical) *Sep. 2017 – July 2021*

Part IA	Computer Science, Materials Science, Mathematics, and Physics.	1st (75.4%)
Part IB	Double Physics and Materials Science.	1st (76.7%)
Part II	Physics: my courses included Advanced Quantum, Astrophysical Fluid Dynamics, Particle and Nuclear Physics, and Relativity.	1st (79.2%)
Part III	Physics: I achieved the highest mark in the cohort in both the Particle Physics and Relativistic Astrophysics and Cosmology major exams. My minor exams included Gauge Field Theory and Advanced Statistical Mechanics.	1st (80.0%)

Beckfoot School, Bingley *Sep. 2010 – June 2017*

A-level	Computer Science: A*, Maths: A*, Further Maths: A*, Physics: A*.
AS-level	Chemistry: A.
GCSE	One Aˆ, eight A*s, two As and one B, including English and maths.

Software and Programming

I have experience with a variety of programming languages and software tools. During my undergraduate degree, I used Java and functional programming in my computer science courses, and Python and Matlab for scientific computing. Through my PhD research, I have worked with a variety of programming languages, primarily Python and Fortran, with some C++. Recently, I have been using GPU acceleration with JAX, and experimenting with Rust. I make habitual use of git for version control of all my projects.

I have made contributions to open-source scientific software, including the widely-used Python libraries pandas and matplotlib, via GitHub. I took the lead in keeping my research group's various packages up to date with the latest Python dependencies, particularly the sample post-processing package anesthetic, which integrates tightly with pandas. I also made improvements to the nested sampling tool PolyChord, which is used by many research groups worldwide.

Research

PhD Project: "Bayesian methods in a flexible universe: sampling, tension and dark energy" *Oct. 2021 – Sep. 2025*
Prof. M.P. Hobson, Dr. W.J. Handley, Prof. A.N. Lasenby

In cosmology, there are many cases where it is useful to reconstruct one–dimensional functions, such as the primordial matter power spectrum or the evolution of the dark energy equation of state. I investigated a free-form approach: a linear spline between a variable number of nodes, using Bayesian statistics to make comparisons between these and other models. A wide range of cosmological datasets were used, including the cosmic microwave background, baryon acoustic oscillations, and Type Ia supernovae. In addition to computing the Bayesian evidence for model comparison, I also used statistics to quantify the possibility of tension between these datasets, and how this tension evolves with model choice.

I also investigated the effects of cluster detection for mode detection and evidence calculation in nested sampling of multi-modal posterior probability distributions.

CDT internship at PolyChord Ltd: MIDAS *Oct. 2023 – March 2024*

During my internship at PolyChord Ltd, I worked on the MIDAS (Maximal Information for Data Acquisition Strategy) project, which uses Bayesian methods to optimise decision making in complex scenarios, where many interdependent decisions are to be made at once. My role in the project was to further develop the existing codebase, focussing on details of the multiprocessing implementation, and addressing assumptions which were inherent in the prior. I also began work on an alternative approach to the problem using Simulation Based Inference.

Ozone damage in broadleaf saplings identified from reflectance spectra *Nov. – Dec. 2023*

Tropospheric ozone is a pollutant which can damage vegetation. Hyperspectral reflectance spectra of leaves can be used to identify signatures of different stressors. I became involved in

this project through a collaboration with Dr. Anna Jones, who had mentioned that the standard analysis pipeline would take many hours to run. I got it down to under a second!

Part III Research Project "Controlling the Beast" *Oct. 2020 – May 2021*

The Magdalena Ridge Observatory Interferometer has a problem with its prototype hardware: the mirrors used to steer light beams from the telescopes to the combining facility are thermally unstable, causing the light beams to wander in angle unacceptably. My project used temperature measurements of the components to understand the dominant causes of this motion, and developed a predictive model to be used to control a corrective system. I also identified issues with the experimental setup which will be corrected in future tests. I learned to use advanced statistical techniques including principal component analysis and singular value decomposition, and machine learning using the Keras API for TensorFlow.

Blundell Lab, University of Cambridge *June 2020 – Sep. 2020*

I remotely investigated a dataset of ultra-deep genome sequences from the Avon Longitudinal Study of Parents and Children (ALSPAC) to create a model for the frequency of sequencing errors in blood samples. I conclusively showed that the position-specific Poisson noise model usually used to identify low-frequency variants is unsuitable. I improved my data analysis skills, in particular in dealing with using files much larger than my laptop's RAM, and gained insight into working in a different field.

Gravitational Physics Group, Cardiff University *July 2019 – Aug. 2019*

I designed and simulated an interferometer using the Finesse software via Python to investigate the holographic principle. Previous setups had 40m arms; the challenge was to fit in a cleanroom only 5m wide while increasing the sensitivity! I also attended meetings discussing the most recent LIGO signals; I enjoyed seeing the collaboration and discussion determining if they were gravitational waves. I shared my own work during these meetings, and the fresh perspectives of the other researchers provided a different approach key to solving my project.

Teaching

- During my PhD, I have supervised Part IA NST Mathematics, for both Queens' and Pembroke colleges. In total, I have supervised 40 students over four years, approximately 367 hours of teaching.
- Through 6th form, I assisted teaching year 7 music classes and after-school small-group sessions with year 10 and 11 students requiring extra help.